

Notes: Peek and Rosengren (AER, 2005)

Unnatural Selection: Perverse Incentives and the Misallocation of Credit in Japan

Contents

1	Big picture	3
2	Incremental contribution	3
3	Data and measurement	4
3.1	Institutional setting: Japanese banks, main banks, and keiretsu	4
3.2	Firm-level and lender-level loan data	4
3.3	Firm health variables	4
3.4	Bank health and exposure variables	4
3.5	Affiliation variables	5
4	Models and empirical specifications	5
4.1	Baseline probability of increased loans	5
4.2	Evergreening hypothesis	6
4.3	Balance-sheet-cosmetics hypothesis	6
4.4	Affiliation hypothesis	6
4.5	Representative-firm predicted probabilities	6
5	Identification and empirical design	6
5.1	Core identification logic	6
5.2	Why lender-level variation matters	7
5.3	Why subsequent stock returns matter	7
5.4	Key threats to interpretation	7
6	Section-by-section study guide	8
7	Tables and figures	9
7.1	Table 1: Sales and Bank Loans for Sogo + Table 2: Firm Debt Composition	9
7.2	Table 3: Loan Increases and the Subsequent Change in Stock Returns + Table 4: Descriptive Statistics for Variables, Market-Traded Bank Sample, 1994–1999	9
7.3	Table 5: Effects of Evergreening and Balance Sheet Cosmetics on Increased Bank Loans + Table 6: Robustness and Bond-Market Split	10
7.4	Table 7: Corporate Affiliations and the Probability of Increased Bank Loans, Part 1 + Table 7 continued, Part 2	11
7.5	Table 8: Factors Affecting Increased Loans by Type of Lender + Table 9: Estimated Probability of Increased Loans for Representative Manufacturing Firm in 1999	12
8	Result map	14

9 Short summary	14
10 Conclusion	14
10.1 Core conclusion	14
10.2 Economic intuition	14
10.3 Incremental contribution	15
10.4 Policy implications	15
10.5 Limitations	15
10.6 Final takeaway	15

1 Big picture

- **Question.** Did Japanese banks during the 1990s allocate credit toward productive firms, or did they systematically extend additional credit to distressed firms in order to hide losses and delay recognition of bad loans?
- **Main mechanism.** Troubled banks had incentives to evergreen loans: lend more to weak borrowers so the borrowers could keep paying interest, avoid bankruptcy, and keep the bank's existing loans from being classified as nonperforming.
- **Context.** Japan's post-bubble banking crisis coincided with regulatory forbearance, understated nonperforming loans, weak loan-loss provisioning, and political pressure to avoid large-scale firm and bank failures.
- **Core empirical object.** The paper studies lender-firm-year data: for each borrowing firm and each lender, did the lender increase loans to that firm from year $t - 1$ to year t ?
- **Main finding 1.** Weak firms were more likely to receive additional bank credit. This is the opposite of a "picking winners" interpretation.
- **Main finding 2.** Evergreening was strongest when the bank had reported capital close to the required capital ratio, supporting the balance-sheet-cosmetics channel.
- **Main finding 3.** Corporate affiliations magnified the behavior: main banks and same-keiretsu lenders were especially likely to support weak borrowers.
- **Main finding 4.** Nonbank lenders without corporate affiliations did not show the same systematic tendency to evergreen weak firms.
- **Economic takeaway.** Bank regulation and relationship lending can preserve credit through downturns, but if regulators permit cosmetic capital accounting, relationship banks may allocate credit to zombie firms and slow aggregate restructuring.

2 Incremental contribution

- **Direct lender-firm evidence of evergreening.** Earlier work documented Japanese banking distress and weak borrower performance, but this paper uses detailed loan links from individual lenders to individual firms to show exactly which lenders increased credit to which borrowers.
- **From bank health to credit misallocation.** The paper moves beyond the question of whether weak banks reduce credit. It shows that weak banks can also *increase* credit to the weakest firms when doing so protects reported capital.
- **Balance-sheet-cosmetics mechanism.** The paper provides a concrete test of whether banks near regulatory capital requirements evergreen more aggressively. This links accounting/regulatory forbearance to real credit allocation.
- **Relationship lending can become distortionary.** Japanese main-bank and keiretsu relationships are often viewed as mechanisms for monitoring and rescue. The paper shows that the same relationships can insulate weak firms from market discipline.
- **Distinguishing banks from nonbanks.** By comparing banks, nonbanks, main banks, secondary banks, same-keiretsu lenders, and government lenders, the paper separates general borrower demand from lender-specific incentives.
- **Lost-decade interpretation.** The results connect micro-level credit allocation to macroeconomic stagnation: banks' attempts to hide losses kept unproductive firms alive and potentially delayed creative destruction.

3 Data and measurement

3.1 Institutional setting: Japanese banks, main banks, and keiretsu

Japanese finance in the 1990s was bank centered. Listed firms often borrowed from multiple banks, and some had especially close relationships with a **main bank**. Many firms were also embedded in **keiretsu** groups, which involved cross-shareholdings, overlapping business ties, and a major bank at the center.

Interpretation: These affiliations are crucial because they can make lending decisions non-arm's-length. The bank may not simply compare expected return and default risk; it may also care about its own balance sheet, group obligations, borrower survival, and regulatory optics.

3.2 Firm-level and lender-level loan data

- The empirical sample uses market-traded Japanese firms over 1994–1999.
- For each firm, the paper observes the lenders from which it borrows and the amount of loans outstanding.
- The key dependent variable is a dummy equal to one if lender j increases loans to firm i from $t - 1$ to t .
- The paper separately examines total loans and long-term loans.
- Observations are at the firm-lender-year level, so the same firm can appear multiple times in a year, once for each lender.

Interpretation: This structure is powerful because it allows the paper to ask not merely whether a distressed firm borrowed more, but *which lender* supplied the new credit.

3.3 Firm health variables

The main firm health variables are lagged one year and include:

- **FROA**: operating income divided by assets;
- **FWORKCAP**: current assets minus current liabilities, divided by assets;
- **FRETLO** and **FRETHI**: indicators or transformations related to very low and very high stock-return performance;
- **FLEV**: leverage;
- **FSALES**: firm sales growth;
- **FLASSET**: firm size;
- bond-market access controls, including whether the firm has bonds outstanding.

Interpretation: The evergreening hypothesis predicts that variables indicating weak firm condition, such as low ROA or low working capital, should increase the likelihood of receiving more bank credit. A cherry-picking story would predict the opposite.

3.4 Bank health and exposure variables

Important bank-side variables include:

- **REQ2**: equals one if the bank’s reported capital ratio is within two percentage points of its required ratio;
- **BRETLO** and **BRETHI**: measures of low and high bank stock-return performance;
- **BEXPOSE**: the lender’s exposure to the firm;
- **MBANK**: equals one if the lender is the firm’s main bank;
- **SAMEK**: equals one if the lender and borrower are in the same keiretsu.

Interpretation: REQ2 is the key balance-sheet-cosmetics variable. If banks near regulatory capital constraints are especially likely to support weak borrowers, then the apparent capital constraint is shaping credit allocation.

3.5 Affiliation variables

- **MBANK** captures main-bank relationships.
- **SAMEK** captures same-keiretsu relationships.
- **KEIR** identifies firms in keiretsu groups.
- **PK** measures the share of ownership held by other keiretsu members among a firm’s top ten equity holders.

Interpretation: Affiliations can mean superior information, but they can also mean soft-budget-constraint support. The empirical question is whether affiliated lenders direct credit to promising firms or shield weak firms from restructuring.

4 Models and empirical specifications

4.1 Baseline probability of increased loans

The baseline empirical model is a random-effects probit:

$$\Pr(LOAN_{i,j,t} = 1) = \Phi(\alpha_0 + \alpha_1 FIRM_{i,t-1} + \alpha_2 BANK_{j,t-1} + \alpha_3 CAPREQ_{j,t-1} + \alpha_4 AFFIL_{i,j,t-1} + \alpha_5 YEAR_t + u_i). \quad (1)$$

- $LOAN_{i,j,t} = 1$ if lender j increases loans to firm i .
- $FIRM$ captures borrower financial health and demand.
- $BANK$ captures lender financial health and exposure.
- $CAPREQ$ captures proximity to required capital ratios.
- $AFFIL$ captures main-bank and keiretsu relationships.
- u_i is a firm random effect.

Interpretation: The random-effects probit uses cross-lender variation for the same firm and cross-firm variation for the same lender. The central test is whether weak firm variables increase loan supply from banks.

4.2 Evergreening hypothesis

The evergreening hypothesis predicts:

$$\frac{\partial \Pr(LOAN_{i,j,t} = 1)}{\partial FROA_{i,t-1}} < 0, \quad \frac{\partial \Pr(LOAN_{i,j,t} = 1)}{\partial WORKCAP_{i,t-1}} < 0. \quad (2)$$

Interpretation: If lower profitability and lower working capital raise the probability of new loans, banks are not allocating credit toward healthy borrowers. They are supporting distressed ones.

4.3 Balance-sheet-cosmetics hypothesis

The balance-sheet-cosmetics hypothesis predicts that the weak-firm effect is stronger for banks close to required capital ratios:

$$\frac{\partial^2 \Pr(LOAN_{i,j,t} = 1)}{\partial REQ2_{j,t-1} \partial WORKCAP_{i,t-1}} < 0. \quad (3)$$

Interpretation: A negative coefficient on interactions like $REQ2 \times WORKCAP$ means that low-working-capital firms are especially likely to receive additional loans from banks near the capital threshold.

4.4 Affiliation hypothesis

The affiliation specification adds main-bank and same-keiretsu interactions:

$$\Pr(LOAN_{i,j,t} = 1) = \Phi(b_0 + b_1 FIRM_{i,t-1} + b_2 BANK_{j,t-1} + b_3 CAPREQ_{j,t-1} + b_4 AFFIL_{i,j,t-1} + b_5 MBANK \times X_1 + b_6) \quad (4)$$

Interpretation: If affiliations are beneficial information channels, affiliated banks should support good firms. If affiliations generate soft budget constraints, affiliated banks should be more likely to support weak firms.

4.5 Representative-firm predicted probabilities

The paper translates probit estimates into predicted probabilities for representative sick and healthy manufacturing firms:

$$\hat{p} = \widehat{\Pr}(LOAN_{i,j,t} = 1 \mid \text{sick or healthy firm, low or high bank exposure, main or secondary bank, same or not same keiretsu}) \quad (5)$$

Interpretation: These predicted probabilities convert coefficient signs into economic magnitudes. They show how much more likely sick firms are to receive additional loans across lender types and capital conditions.

5 Identification and empirical design

5.1 Core identification logic

The paper does not rely on a randomized experiment. Its empirical design is diagnostic: it compares loan increases across lender-firm pairs, conditioning on borrower health, lender health, capital constraints, affiliations, and year effects.

- If banks are picking winners, healthier firms should receive more loans.
- If banks are evergreening, weaker firms should receive more loans.
- If capital cosmetics matter, this pattern should intensify for banks close to minimum capital requirements.
- If affiliations matter, main-bank and same-keiretsu relationships should amplify the support of weak firms.

Interpretation: The research design is strongest because the hypotheses make sharp sign predictions that differ from standard efficient lending.

5.2 Why lender-level variation matters

A firm may demand more credit because it is growing, or because it is distressed. A bank may supply more credit because it sees good prospects, or because it wants to hide losses. Observing firm-lender pairs helps separate these stories.

- The same firm can borrow from main banks, secondary banks, same-keiretsu lenders, non-keiretsu lenders, nonbanks, and government lenders.
- If all lenders increase credit similarly, demand may dominate.
- If only banks with incentives to evergreen increase credit, lender incentives are central.

Interpretation: This is why the contrast between affiliated banks and nonaffiliated nonbanks is important. It helps rule out a pure borrower-demand story.

5.3 Why subsequent stock returns matter

The paper checks whether firms receiving increased bank loans subsequently perform well or poorly. If banks are cherry-picking future winners, loan recipients should overperform. If banks are evergreening, loan recipients should underperform.

Interpretation: The subsequent-return tests support evergreening: firms receiving increased loans in the later period are disproportionately in the lowest stock-return quartile.

5.4 Key threats to interpretation

- **Loan demand.** Weak firms may request more loans, so coefficients combine demand and supply. The paper addresses this by comparing lender types and affiliations.
- **Unobserved private information.** Banks may know something positive about weak firms. Subsequent returns and poor performance help reject this interpretation.
- **Survivorship and listed-firm sample.** The sample is market-traded firms, not all firms. The results are therefore about visible listed borrowers during the crisis.
- **Regulatory forbearance not directly observed.** The paper infers balance-sheet-cosmetics incentives from capital-ratio proximity and lending patterns, not from internal regulator documents.

Interpretation: The empirical evidence is not a single causal coefficient. It is a set of mutually reinforcing patterns: weak firms get more loans, constrained banks do more of this, affiliations amplify it, and nonaffiliated nonbanks do not behave the same way.

6 Section-by-section study guide

- **Introduction.** Defines evergreening and motivates the Japanese lost decade as a case of credit misallocation.
- **Section I.** Explains Japanese bank-firm relationships, main banks, keiretsu affiliations, and the Sogo example.
- **Section II.** Develops the three hypotheses: evergreening, balance-sheet cosmetics, and affiliation effects.
- **Section III.** Describes firm-bank-year data and estimates probit models of loan increases.
- **Tables 1–3.** Establish motivating evidence that bank credit shifted toward weak firms rather than future winners.
- **Tables 4–9.** Provide the formal regression evidence and economic magnitudes.
- **Conclusion.** Links micro-credit allocation to macro stagnation and the delay of restructuring.

7 Tables and figures

7.1 Table 1: Sales and Bank Loans for Sogo + Table 2: Firm Debt Composition

Read:

- Table 1 shows that Sogo's sales growth turned negative in 1992 and remained weak, yet bank loans continued to rise in many years.
- Sogo's main bank share rose from 7.25 percent in 1990 to 34.51 percent in 1999, even as the firm deteriorated.
- This case study illustrates evergreening: banks expanded exposure to a failing borrower rather than forcing restructuring.
- Table 2 shows that Japanese firms increased reliance on total loans and bank loans relative to bonds during the 1990s.
- The shift is especially strong for low-ROA firms, which move heavily away from bonds and toward bank loans.

Detailed interpretation: Table 1 gives a concrete micro example of the paper's mechanism. Table 2 shows that the Sogo episode was not simply anecdotal: the broader sample exhibits a systematic move away from arm's-length bond finance toward relationship bank finance, especially among weak firms.

Economic message: Banks did not just fail to cut off credit to weak firms; they often increased support precisely when market discipline should have tightened.

TABLE 1—SALES AND BANK LOANS FOR SOGO

	Sales growth	Bank loan growth	Main bank share	Share of second and third banks	Share of other banks
1990	11.68	3.76	7.25	13.16	79.59
1991	7.12	14.37	6.34	11.51	82.15
1992	-0.92	18.37	5.35	9.72	84.92
1993	-9.83	15.43	4.64	8.42	86.94
1994	-9.18	18.86	10.47	11.05	78.48
1995	-8.53	44.35	21.59	17.02	61.38
1996	-34.51	-1.65	23.89	16.04	60.06
1997	16.80	4.24	27.45	16.45	56.10
1998	-5.38	-0.02	27.44	16.44	56.11
1999	-8.28	12.20	34.51	15.93	49.56
2000	-6.16	-13.40	27.32	15.75	56.93

Table 1: Sales and Bank Loans for Sogo

TABLE 2—FIRM DEBT COMPOSITION
(Mean of values for individual firms; measured as a percent of bonds plus total loans)

Panel A	All firms				Firms with bonds outstanding			
	Bonds	Total loans	Bank loans	Main bank loans	Bonds	Total loans	Bank loans	Main bank loans
1993	27.51	72.49	55.65	16.92	38.85	61.15	46.78	13.36
1994	27.68	72.32	55.39	16.96	38.06	61.94	45.25	12.94
1995	26.07	73.93	56.82	17.53	37.35	62.65	46.41	13.07
1996	22.96	77.04	59.68	17.70	36.26	63.74	47.56	12.86
1997	21.76	78.24	61.00	19.21	35.89	64.11	47.75	14.08
1998	20.06	79.94	61.69	20.06	34.41	65.59	47.31	14.27
1999	19.62	80.38	62.44	20.92	35.31	64.69	45.99	14.51

Panel B: Firms with bonds outstanding	Firms in lowest quartile of ROA				Firms in top quartile of ROA			
	Bonds	Total loans	Bank loans	Main bank loans	Bonds	Total loans	Bank loans	Main bank loans
1993	20.54	79.46	58.64	17.48	49.14	50.86	39.00	12.02
1994	18.72	81.28	61.85	19.14	52.89	47.11	35.66	9.92
1995	16.25	83.75	63.95	21.07	56.30	43.70	31.55	8.50
1996	13.24	86.76	66.10	21.99	48.07	51.93	40.03	9.27
1997	11.09	88.91	68.65	23.34	48.17	51.83	37.75	11.60
1998	9.12	90.88	70.47	23.66	44.09	55.91	39.58	12.42
1999	8.84	90.16	71.77	24.77	47.03	52.97	40.36	12.60

Table 2: Firm Debt Composition

7.2 Table 3: Loan Increases and the Subsequent Change in Stock Returns + Table 4: Descriptive Statistics for Variables, Market-Traded Bank Sample, 1994–1999

Read:

- Table 3 asks whether firms receiving increased bank credit subsequently do well.
- In 1996–1998, firms receiving increased total bank loans are more concentrated in the lowest subsequent stock-return quartile.
- This pattern is inconsistent with banks cherry-picking strong firms.
- Table 4 shows the summary statistics: the mean probability of total loans increasing is 0.235, while the mean probability of long-term loans increasing is 0.133.

- The sample contains wide variation in firm health, bank exposure, main-bank status, and same-keiretsu relationships.

Detailed interpretation: Table 3 is a key falsification of the “banks knew better” interpretation. If relationship banks had superior positive information, the firms they supported should have outperformed. Instead, later performance is weak. Table 4 gives the variable distribution that underlies the probit tests.

Economic message: Additional bank credit flows to firms that subsequently perform poorly, pointing to loss-hiding and support motives rather than efficient credit allocation.

TABLE 4—DESCRIPTIVE STATISTICS FOR VARIABLES,
MARKET-TRADED BANK SAMPLE, 1994–1999

	Mean	Std. dev.	Min	Max
Total loans up	0.235	0.424	0	1
Long-term loans up	0.133	0.339	0	1
FROA	2.970	3.292	−27.291	26.193
FWORKCAP	10.283	17.431	−133.592	89.354
FSALES	0.627	11.355	−65.281	249.673
FLEV	69.804	17.759	3.065	189.296
FLASSET	4.366	1.606	−0.242	8.860
FENBMKT	0.026	0.158	0	1
FINBMKT	0.598	0.490	0	1
FEXBMKT	0.051	0.220	0	1
REQ2	0.714	0.452	0	1
BEXPOSE	5.870	9.877	0	100
MBANK	0.065	0.247	0	1
MBEXPOSE	25.383	14.672	0	100
KEIR	0.466	0.499	0	1
PK	11.211	16.428	0	88.300
SAMEK	0.057	0.232	0	1

TABLE 3—LOAN INCREASES AND THE SUBSEQUENT CHANGE IN STOCK RETURNS

Loan category	Period	Quartile of stock returns			
		1	2	3	4
Total bank loans	1994–1995	26.32	24.30	24.30	25.08
	1996–1998	28.58	25.41	23.65	22.36
Main bank same	1994–1995	23.46	22.31	31.92	22.31
Keiretsu	1996–1998	28.73	25.79	24.21	21.27
Main bank not	1994–1995	25.05	26.10	21.50	27.35
Same keiretsu	1996–1998	28.94	25.08	23.47	22.51
Secondary bank	1994–1995	21.37	20.61	32.06	25.95
Same keiretsu	1996–1998	30.63	26.80	23.20	19.37
Secondary bank	1994–1995	25.72	23.88	24.23	26.18
Not same keiretsu	1996–1998	29.02	25.50	23.35	22.13

Notes: Each row considers the set of firms that obtained an increase in that category of loans during the year. The number in the cell indicates the percentage of those firms whose stock return in the subsequent year placed them in each stock return quartile.

Table 3: Loan Increases and Subsequent Stock Returns

Table 4: Descriptive Statistics

7.3 Table 5: Effects of Evergreening and Balance Sheet Cosmetics on Increased Bank Loans + Table 6: Robustness and Bond-Market Split

Read:

- Table 5 is the baseline random-effects probit. FROA and FWORKCAP are negative and significant, showing that weak firms are more likely to receive increased loans.
- REQ2 interactions with weak-firm variables support the balance-sheet-cosmetics channel: banks near capital requirements increase loans more to troubled firms.
- BEXPOSE is positive, consistent with banks with larger preexisting exposure having stronger incentives to keep borrowers alive.
- Table 6 shows robustness for long-term loans and separate samples of firms with and without bonds outstanding.
- The evergreening pattern is particularly strong for firms with bonds outstanding, suggesting that firms priced out of arm’s-length markets return to relationship banks.

Detailed interpretation: Table 5 is the paper’s central empirical result. Negative coefficients on weak firm health variables mean that weaker firms get more bank credit, not less. Table 6 shows that this result is not an artifact of total-loan measurement and that the bond-market split reinforces the idea of firms migrating back to banks when arm’s-length finance becomes costly.

Economic message: Troubled banks appear to allocate credit to protect their own reported balance sheets, and the credit flows toward firms least likely to use it productively.

TABLE 5—THE EFFECTS OF EVERGREENING AND BALANCE SHEET COSMETICS ON INCREASED BANK LOANS
(Random effects probit specification)

	Full sample	Loans _{<i>i,t</i>-1} > 0	
		All observations	Extreme observations eliminated
FROA	-0.0075** (0.0028)	-0.0085** (0.0031)	-0.0102** (0.0032)
FWORKCAP	-0.0097** (0.0006)	-0.0120** (0.0007)	-0.0114** (0.0007)
FRETLO	-0.0392** (0.0064)	-0.0273** (0.0070)	-0.0280** (0.0071)
FRETHI	-0.0624** (0.0060)	-0.0645** (0.0066)	-0.0656** (0.0067)
FLEV	-0.0094** (0.0005)	-0.0140** (0.0006)	-0.0127** (0.0007)
FSALES	0.0024** (0.0005)	0.0020** (0.0003)	0.0022** (0.0004)
REQ2	0.0582** (0.0157)	0.0223 (0.0177)	0.0218 (0.0182)
REQ2*FROA	-0.0095* (0.0034)	-0.0080* (0.0039)	-0.0083* (0.0041)
REQ2*WORKCAP	-0.0030** (0.0006)	-0.0032** (0.0007)	-0.0032** (0.0007)
BRETLO	-0.1163** (0.0126)	-0.0682** (0.0139)	-0.0676** (0.0142)
BRETHI	-0.0697** (0.0133)	-0.0317* (0.0149)	-0.0285 (0.0150)
BEXPOSE	0.0180** (0.0006)	0.0191** (0.0007)	0.0211** (0.0008)
MBANK	-0.0601 (0.0326)	-0.0732* (0.0348)	-0.1137** (0.0354)
MBRETLO	-0.0209** (0.0075)	-0.0266** (0.0082)	-0.0262** (0.0083)
MBRETHI	0.0529** (0.0081)	0.0464** (0.0087)	0.0535** (0.0088)
MBEXPOSE	-0.0015** (0.0004)	-0.0061** (0.0005)	-0.0039** (0.0006)
KEIR	0.1389** (0.0222)	0.1160** (0.0254)	0.1324** (0.0261)
PK	-0.0050** (0.0008)	-0.0035** (0.0008)	-0.0037** (0.0009)
SAMEK	0.1831** (0.0234)	0.1581** (0.0244)	0.1499** (0.0248)
Number of observations	95,566	78,028	76,696
Number of firms	1,215	1,185	1,169
Mean of dependent variable	0.2383	0.2549	0.2552
Correctly predicted (percent)	99.43	99.25	99.10
Average likelihood (percent)	61.83	60.70	59.84

Notes: The estimated equations also include FLASSET, FENBMKT, FINBMKT, FEXBMKT, a set of industry dummy variables, a set of annual dummy variables, and a set of bank-type

Table 5: Evergreening and Balance-Sheet Cosmetics

TABLE 6—THE EFFECTS OF EVERGREENING AND BALANCE SHEET COSMETICS ON INCREASED BANK LOANS
(Random effects probit specification: Loans_{*i,t*-1} > 0)

	Long-term loans full sample	All loans Bonds _{<i>i,t</i>-1} = 0	All loans Bonds _{<i>i,t</i>-1} > 0
FROA	-0.0162** (0.0039)	-0.0010 (0.0055)	-0.0132** (0.0039)
FWORKCAP	-0.0166** (0.0008)	-0.0083** (0.0014)	-0.0145** (0.0009)
FRETLO	-0.0141 (0.0088)	-0.0007 (0.0149)	-0.0231** (0.0090)
FRETHI	-0.0667** (0.0086)	0.0431** (0.0152)	-0.1203** (0.0078)
FLEV	-0.0092** (0.0007)	-0.0111** (0.0014)	-0.0077** (0.0007)
FSALES	0.0016** (0.0004)	0.0027** (0.0006)	0.0003 (0.0004)
REQ2	0.0161 (0.0210)	-0.0073 (0.0314)	0.0454** (0.0221)
REQ2*FROA	-0.0063 (0.0045)	-0.0042 (0.0069)	-0.0143** (0.0051)
REQ2*WORKCAP	-0.0041** (0.0008)	-0.0019 (0.0013)	-0.0033** (0.0009)
BRETLO	-0.0460** (0.0172)	-0.0974** (0.0249)	-0.0580** (0.0173)
BRETHI	-0.0660** (0.0187)	-0.0476 (0.0265)	-0.0247 (0.0182)
BEXPOSE	0.0257** (0.0007)	0.0183** (0.0012)	0.0190** (0.0010)
MBANK	-0.1328** (0.0327)	-0.0359 (0.0551)	-0.0892 (0.0478)
MBRETLO	-0.0524** (0.0104)	-0.0219 (0.0191)	-0.0359** (0.0101)
MBRETHI	0.0224** (0.0107)	0.0956** (0.0161)	0.0143 (0.0118)
MBEXPOSE	-0.0084** (0.0006)	-0.0089** (0.0011)	-0.0061** (0.0007)
KEIR	0.1792** (0.0354)	0.0049 (0.0658)	0.0540 (0.0314)
PK	-0.0091** (0.0011)	-0.0016 (0.0019)	0.0007 (0.0011)
SAMEK	0.1437** (0.0259)	0.1603** (0.0457)	0.1583** (0.0306)
Number of observations	78,028	26,202	51,826
Number of firms	1,185	686	845
Mean of dependent variable	0.1512	0.2753	0.2446
Correctly predicted (percent)	99.62	98.71	99.04
Average likelihood (percent)	69.18	58.03	60.90

Notes: The estimated equations also include FLASSET, FENBMKT, FINBMKT, FEXBMKT, a set of industry dummy variables, a set of annual dummy variables, and a set of bank-type dummy variables. Standard errors are in parentheses.
* Significant at the 5-percent level; ** significant at the 1-percent level.

Table 6: Robustness and Bond-Market Split

7.4 Table 7: Corporate Affiliations and the Probability of Increased Bank Loans, Part 1 + Table 7 continued, Part 2

Read:

- Table 7 adds main-bank and same-keiretsu interactions.
- MBANK is strongly positive: main banks are more likely to increase loans.
- SAMEK is positive in many specifications, showing that same-keiretsu ties increase support.
- Interactions show that affiliations change the intensity and distribution of evergreening.
- The continued portion of Table 7 helps separate main-bank effects from same-keiretsu effects and bank exposure effects.

Detailed interpretation: Table 7 shows that Japanese corporate affiliations are not merely information channels. They can also be bailout channels. Main banks and same-keiretsu lenders appear to support troubled firms more than arm's-length lenders, consistent with soft budget constraints.

Economic message: Relationship finance has a dark side: the same ties that facilitate monitoring can also delay restructuring when banks and firms are jointly distressed.

TABLE 7—CORPORATE AFFILIATIONS AND THE PROBABILITY OF INCREASED BANK LOANS
(Random effects probit specification: $Loans_{i,j,t-1} > 0$)

	Full sample	Extreme observations eliminated	Bonds _{i,j,t-1} = 0	Bonds _{i,j,t-1} > 0
FROA	-0.0099** (0.0032)	-0.0103** (0.0035)	-0.0006 (0.0059)	-0.0112** (0.0041)
FWORKCAP	-0.0124** (0.0007)	-0.0124** (0.0007)	-0.0063** (0.0014)	-0.0149** (0.0010)
FRETLO	-0.0488** (0.0089)	-0.0405** (0.0091)	-0.0225 (0.0187)	-0.0321** (0.0115)
FRETHI	-0.0634** (0.0082)	-0.0658** (0.0082)	0.0324 (0.0195)	-0.1069** (0.0097)
FLEV	-0.0145** (0.0007)	-0.0129** (0.0006)	-0.0074** (0.0013)	-0.0083** (0.0008)
FSALES	0.0020** (0.0003)	0.0022** (0.0004)	0.0026** (0.0006)	0.0003 (0.0004)
REQ2	0.0103 (0.0186)	0.0089 (0.0190)	-0.0141 (0.0333)	0.0338 (0.0235)
REQ2*FROA	-0.0066 (0.0041)	-0.0065 (0.0043)	-0.0035 (0.0072)	-0.0134** (0.0053)
REQ2*FWORKCAP	-0.0031** (0.0007)	-0.0033** (0.0008)	-0.0019 (0.0014)	-0.0034** (0.0009)
BRETLO	-0.0804** (0.0152)	-0.0798** (0.0151)	-0.1162** (0.0265)	-0.0670** (0.0190)
BRETHI	-0.0270 (0.0161)	-0.0249 (0.0161)	-0.0499 (0.0292)	-0.0186 (0.0199)
BEXPOSE	-0.0311** (0.0009)	-0.0348** (0.0010)	0.0301** (0.0015)	0.0324** (0.0012)
MBRETLO	-0.0223* (0.0100)	-0.0236* (0.0102)	-0.0180 (0.0222)	-0.0330** (0.0121)
MBRETHI	0.0486** (0.0101)	0.0485** (0.0102)	0.0985** (0.0192)	0.0195 (0.0136)
MBEXPOSE	-0.0012* (0.0006)	0.0002 (0.0006)	-0.0053** (0.0013)	-0.0004 (0.0008)
KEIR	0.0564* (0.0266)	0.1568** (0.0260)	0.0282 (0.0679)	0.1092** (0.0337)
PK	-0.0006 (0.0009)	-0.0037** (0.0010)	-0.0043* (0.0020)	0.0008 (0.0012)
MBANK	0.5963** (0.0998)	0.5962** (0.1061)	0.6154** (0.1597)	0.5983** (0.1409)
MBANK*FROA	-0.0202** (0.0066)	-0.0217** (0.0080)	-0.0197 (0.0101)	-0.0185 (0.0096)
MBANK*FWORKCAP	0.0006 (0.0013)	0.0001 (0.0014)	-0.0002 (0.0020)	0.0014 (0.0019)
MBANK*FRETLO	0.0472 (0.0533)	0.0469 (0.0540)	0.0654 (0.0852)	0.0322 (0.0732)
MBANK*FRETHI	0.0322 (0.0545)	0.0247 (0.0545)	0.0339 (0.0929)	0.0053 (0.0707)
MBANK*REQ2	-0.0170 (0.0659)	0.0013 (0.0664)	-0.0298 (0.0921)	0.0019 (0.1084)
MBANK*BRETLO	0.1140* (0.0576)	0.1258* (0.0580)	0.0523 (0.0902)	0.1501 (0.0793)
MBANK*BRETHI	-0.0464 (0.0636)	-0.0430 (0.0649)	-0.0489 (0.1034)	-0.0265 (0.0835)
MBANK*BEXPOSE	-0.0292** (0.0017)	-0.0326** (0.0019)	-0.0273** (0.0028)	-0.0314** (0.0024)
MBANK*KEIR	-0.0117 (0.0138)	-0.0120 (0.0139)	0.0109 (0.0239)	-0.0220 (0.0192)
MBANK*PK	0.0029 (0.0019)	0.0031 (0.0018)	0.0023 (0.0030)	0.0016 (0.0029)
MBANK*SK	0.4162** (0.1604)	0.4619** (0.1596)	0.2513 (0.2616)	0.5111* (0.2335)

TABLE 7—(CONTINUED)

	Full sample	Extreme observations eliminated	Bonds _{i,j,t-1} = 0	Bonds _{i,j,t-1} > 0
SAMEK	0.1815* (0.0879)	0.1793* (0.0887)	-0.1004 (0.1859)	0.3219** (0.1113)
SK*FROA	0.0066 (0.0081)	0.0029 (0.0091)	0.0230 (0.0128)	-0.0097 (0.0119)
SK*FWORKCAP	-0.0016 (0.0017)	-0.0015 (0.0017)	-0.0002 (0.0030)	-0.0013 (0.0024)
SK*FRETLO	0.1340** (0.0479)	0.1273* (0.0487)	0.1326 (0.0855)	0.1357* (0.0633)
SK*FRETHI	-0.0391 (0.0468)	-0.0441 (0.0471)	0.0844 (0.0958)	-0.0929 (0.0591)
SK*REQ2	0.1107* (0.0570)	0.1144* (0.0577)	0.1021 (0.1005)	0.1426 (0.0792)
SK*MBANK*REQ2	-0.3993** (0.1480)	-0.4252** (0.1465)	-0.2389 (0.2390)	-0.4751* (0.2177)
SK*BRETLO	0.0176 (0.0614)	0.0146 (0.0624)	0.1489 (0.1130)	-0.0292 (0.0794)
SK*BRETHI	-0.0192 (0.0665)	-0.0326 (0.0679)	0.0795 (0.1256)	-0.0652 (0.0839)
SK*BEXPOSE	-0.0098** (0.0028)	-0.0106** (0.0029)	-0.0090 (0.0049)	-0.0097** (0.0037)
SK*MBRETLO	-0.0401 (0.0514)	-0.0404 (0.0519)	-0.0400 (0.0947)	-0.0588 (0.0665)
SK*MBRETHI	-0.1205* (0.0596)	-0.1220* (0.0611)	-0.1516 (0.1026)	-0.1125 (0.0798)
SK*MBEXPOSE	0.0030 (0.0021)	0.0036 (0.0022)	0.0051 (0.0042)	0.0017 (0.0027)
SK*PK	-0.0030 (0.0016)	-0.0031 (0.0016)	-0.0007 (0.0028)	-0.0048* (0.0023)
Number of observations	78,028	76,696	26,202	51,826
Number of firms	1,185	1,169	686	845
Mean of dependent variable	0.2549	0.2552	0.2753	0.2446
Correctly predicted (percent)	98.97	99.11	98.70	98.86
Average likelihood (percent)	60.41	60.95	58.36	61.49

Notes: The estimated equations also include FLASSET, FENBMKT, FINBMKT, FEXBMKT, a set of industry dummy variables, a set of annual dummy variables, and a set of bank-type dummy variables. Standard errors are in parentheses. * Significant at the 5-percent level; ** significant at the 1-percent level.

Table 7: Corporate Affiliations, Part 1

Table 7 continued, Part 2

7.5 Table 8: Factors Affecting Increased Loans by Type of Lender + Table 9: Estimated Probability of Increased Loans for Representative Manufacturing Firm in 1999

Read:

- Table 8 separates lender types: main banks, secondary banks, nonbanks, same-keiretsu and non-same-keiretsu lenders, and government lenders.
- Nonaffiliated nonbanks do not show the same systematic evergreening incentives as banks.
- Government and affiliated lenders show patterns more consistent with support for weak firms.
- Table 9 translates coefficients into predicted probabilities for sick and healthy firms.
- Sick firms have substantially higher probabilities of receiving increased loans than healthy firms across many lender categories.
- The sick-healthy gap is large for main banks and same-keiretsu lenders, especially under capital pressure.

Detailed interpretation: Table 8 is important because it distinguishes lender-side incentives from pure borrower demand. If weak firms simply demanded more credit, nonbanks should behave similarly. They do not. Table 9 shows economic magnitudes: evergreening is not only statistically significant but quantitatively large.

Economic message: The misallocation of credit is concentrated among lenders with the strongest incentives or obligations to preserve troubled borrowers.

TABLE 8—FACTORS AFFECTING THE PROBABILITY OF INCREASED LOANS, BY TYPE OF LENDER
(Random effects probit specification: $Loans_{i,j,t} > 0$)

	Main bank Same K	Main bank Not same K	Secondary Same K	Secondary Not same K	Nonbank Same K	Nonbank Not same K	Government
Intercept	0.5779** (0.1489)	0.4100** (0.1159)	0.1078 (0.1273)	-0.2766** (0.0854)	0.1161 (0.1496)	-0.2988** (0.0891)	-0.3972** (0.0932)
FROA	-0.0119 (0.0127)	-0.0448** (0.0065)	-0.0097 (0.0098)	-0.0187** (0.0016)	-0.0045 (0.0131)	0.0067* (0.0028)	0.0009 (0.0045)
FWORKCAP	-0.0163** (0.0027)	-0.0136** (0.0013)	-0.0162** (0.0021)	-0.0139** (0.0004)	-0.0175** (0.0026)	-0.0103** (0.0006)	-0.0163** (0.0008)
FRETLO	0.1417 (0.0750)	0.0227 (0.0557)	0.1000 (0.0571)	-0.0341** (0.0088)	-0.0033 (0.0771)	-0.1132** (0.0206)	-0.0498 (0.0325)
FRETHI	-0.1250 (0.0779)	-0.0051 (0.0574)	-0.0818 (0.0606)	-0.0803** (0.0082)	-0.0703 (0.0767)	-0.1031** (0.0177)	-0.0788* (0.0318)
BEXPOSE	0.0014 (0.0024)	0.0024 (0.0016)	0.0241** (0.0040)	0.0388** (0.0008)	0.0052 (0.0059)	0.0041 (0.0022)	0.0167** (0.0010)
MBRETLO	-0.1476 (0.0824)	-0.0933 (0.0658)	-0.0261 (0.0646)	-0.0446** (0.0096)	0.0675 (0.0850)	0.1062** (0.0228)	0.1076** (0.0358)
MBRETHI	-0.0013 (0.0820)	0.0402 (0.0588)	-0.0411 (0.0609)	-0.0029 (0.0097)	0.2023** (0.0758)	0.0702** (0.0206)	0.0963** (0.0347)
MBEXPOSE			0.0004 (0.0022)	0.0008** (0.0006)	-0.0016 (0.0034)	-0.0028** (0.0010)	-0.0063** (0.0012)
KEIR		0.1010 (0.1386)		0.0779** (0.0221)		0.2881** (0.0313)	0.3462** (0.0366)
PK	-0.0026 (0.0024)	-0.0014 (0.0042)	-0.0051** (0.0017)	-0.0023** (0.0008)	-0.0064** (0.0021)	-0.0071** (0.0013)	-0.0087** (0.0013)
Number of observations				111,577			
Number of firms				1,200			
Mean of dependent variable				0.2423			
Correctly predicted (percent)				99.00			
Average likelihood (percent)				61.31			

Notes: The estimated equations also include FLASSET, FENBMKT, FINBMKT, FEXBMKT, FLEV, FSALES, a set of industry dummy variables, and a set of annual dummy variables. Standard errors are in parentheses. * Significant at the 5-percent level; ** significant at the 1-percent level.

Table 8: Increased Loans by Type of Lender

TABLE 9—ESTIMATED PROBABILITY OF INCREASED LOANS FOR REPRESENTATIVE MANUFACTURING FIRM IN 1999

		Low bank exposure			High bank exposure		
		Sick firm	Healthy firm	Difference	Sick firm	Healthy firm	Difference
Panel A: REQ2 = 1	Main Bank	Not in K	44.9	24.5	20.4	45.3	24.8
		Same K	55.3	28.0	27.3	51.8	25.1
	Secondary Bank	Not in K	28.3	14.2	14.1	32.3	16.9
		Same K	42.0	19.6	22.4	45.1	21.9
Panel B: REQ2 = 0		Low bank exposure			High bank exposure		
		Sick firm	Healthy firm	Difference	Sick firm	Healthy firm	Difference
Main bank	Not in K	45.6	24.5	21.1	46.0	32.2	13.8
	Same K	65.2	45.0	20.2	61.9	41.5	20.3
	Secondary bank	Not in K	28.5	19.3	9.2	32.5	22.6
	Same K	38.6	22.9	15.7	41.7	25.4	16.3

Note: See text for a detailed description of the calculations.

Table 9: Predicted Probabilities for Representative Firm

8 Result map

Result 1: Evergreening

Weak firms, measured by low ROA and low working capital, are more likely to receive increased loans. This contradicts efficient credit allocation and supports the idea that banks roll over problem credit.

Result 3: Affiliation effects

Main-bank and same-keiretsu relationships amplify support for weak firms, showing that relationship lending can become a soft-budget-constraint mechanism.

Result 2: Balance-sheet cosmetics

Banks near required capital ratios are especially likely to support weak firms, consistent with incentives to keep loans current and avoid recognizing nonperforming assets.

Result 4: Nonbank contrast

Nonaffiliated nonbanks do not exhibit the same behavior, helping distinguish bank evergreening incentives from borrower loan demand.

Result 5: Macro implication

Credit misallocation preserves weak firms and delays exit or restructuring. The paper therefore links micro-level bank incentives to Japan's slow recovery after the asset-price collapse.

9 Short summary

- The paper studies why Japanese banks in the 1990s continued lending to distressed firms.
- The key mechanism is evergreening: new loans let weak firms keep paying interest and avoid bankruptcy, allowing banks to avoid recognizing nonperforming loans.
- The paper uses lender-firm-year loan data, which allows a direct test of lender incentives.
- Weak firms are more likely to receive additional bank loans.
- Firms receiving increased bank loans subsequently perform poorly, undermining the “picking winners” interpretation.
- Banks near required capital ratios evergreen more aggressively, supporting the balance-sheet-cosmetics hypothesis.
- Main-bank and same-keiretsu affiliations amplify the pattern.
- Nonaffiliated nonbanks do not behave like banks, indicating that lender incentives matter.
- The paper connects bank forbearance, government pressure, accounting opacity, and corporate affiliations to credit misallocation.
- The broader implication is that financial-sector forbearance can slow restructuring and prolong stagnation.

10 Conclusion

10.1 Core conclusion

Peek and Rosengren show that Japanese banks in the 1990s often allocated additional credit to financially weak borrowers rather than to firms with the best prospects. This lending behavior is consistent with evergreening: banks extended credit to keep troubled borrowers alive and avoid recognizing losses on their own balance sheets.

10.2 Economic intuition

The intuition is a soft-budget-constraint problem created by bank fragility and regulatory forbearance. When a weak borrower fails, the bank must recognize loan losses. If the bank is already near its regulatory capital

requirement, this recognition can force costly corrective action. Therefore, a troubled bank may prefer to lend just enough to keep the borrower current, even if the borrower is not economically viable. This is individually rational for the bank but socially costly.

10.3 Incremental contribution

The paper's incremental contribution is to show, with detailed lender-firm data, that Japan's banking crisis was not only a story of reduced credit supply. It was also a story of distorted credit supply. Banks did not merely fail to lend enough; they lent too much to the wrong firms. By connecting firm distress, bank capital pressure, lender exposure, and affiliation ties, the paper provides a micro-level mechanism for Japan's macroeconomic stagnation.

10.4 Policy implications

The paper implies that bank recapitalization and loan-loss recognition are central to recovery. If regulators allow weak banks to hide losses, banks may preserve zombie borrowers and crowd out more productive firms. Relationship banking can be valuable in normal times, but in a crisis it can become a channel through which losses are delayed and misallocation is preserved.

10.5 Limitations

The design is observational and cannot randomly assign bank capital constraints or borrower distress. Loan demand and loan supply are difficult to fully separate. The sample is limited to listed firms and observed lender relationships. Still, the consistency of the evidence across subsequent returns, bank capital pressure, affiliations, and lender-type comparisons makes the evergreening interpretation persuasive.

10.6 Final takeaway

The enduring lesson is that forbearance is not neutral. When banks are weak and supervision tolerates accounting opacity, banks may use new lending to postpone the recognition of old losses. This can prevent immediate collapse but prolongs stagnation by keeping unproductive firms alive and delaying the reallocation of capital.